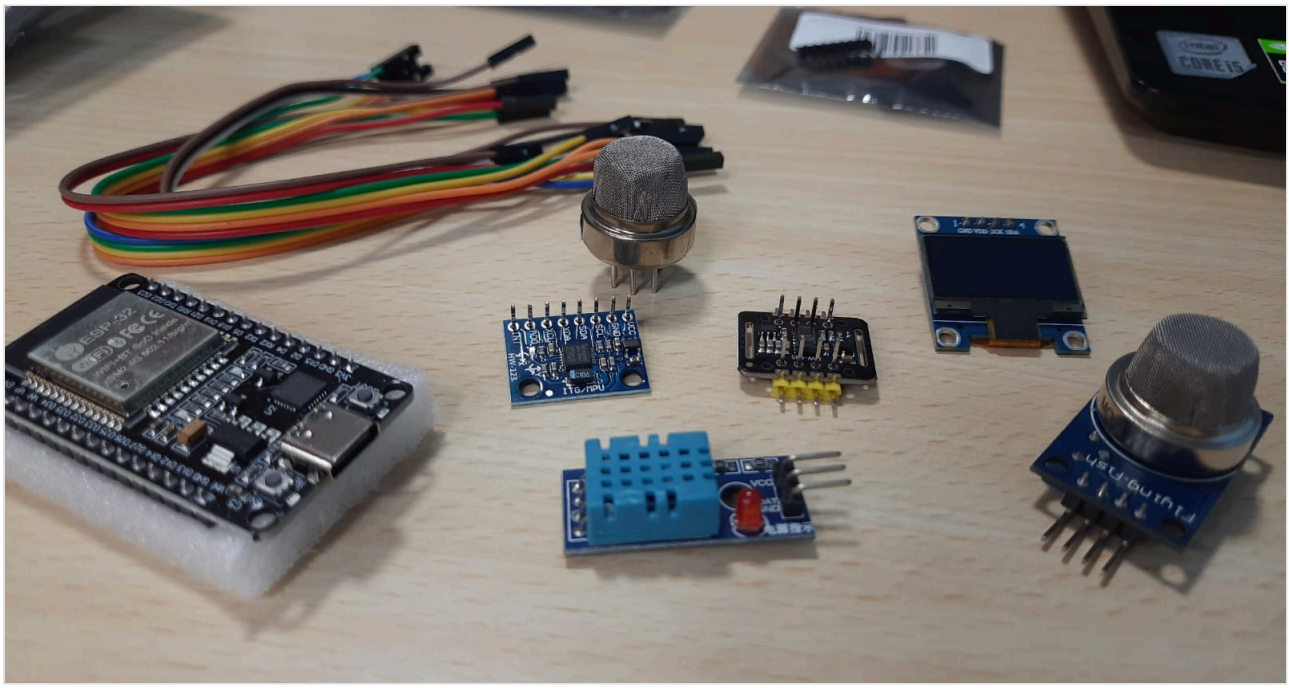
EMBEDDED SYSTEMS & SENSOR TELEMETRY SERIES | TECHNICAL REPORT

Multi-Sensor Driver Monitoring & Telemetry Test Rig

ESP32 Integration with MPU-6050 & MAX30102: Hardware Pin Mapping, Shared I²C Architecture, and Industrial Operator Telemetry



LinkedIn Github

HOST ARCHITECTURE
ESP32 DevKit (30-pin)

COMMUNICATION BUS
Shared Hardware I²C (400 kHz)

SENSORS CONFIGURED
MPU-6050 & MAX30102

PRIMARY APPLICATION
Operator Fatigue & Telemetry

Hardware Setup, Pinout Topology & Bus Arbitration

Detailed Circuit Interconnection and Shared Bus Timing Specifications

1. System Architecture & Integration Objective

This test rig integrates an **MPU-6050** (6-axis inertial motion tracker) and a **MAX30102** (pulse oximetry and heart rate monitor) onto a shared **I²C bus** driven by an **ESP32 microcontroller**. By pairing real-time kinematic posture sensing (head tilt, nodding, sudden jerks) with physiological photoplethysmogram telemetry (heart rate variability and arterial oxygen saturation), the system forms a unified edge telemetry platform for operator alertness, micro-sleep detection, and fatigue tracking.

The architecture consolidates all communication onto the primary hardware I²C interface (GPIO21/GPIO22), leveraging independent device addressing to achieve multi-modal streaming without requiring external multiplexers, auxiliary buses, or complex interrupt topologies.

2. Hardware Pinout & Breadboard Wiring Topology

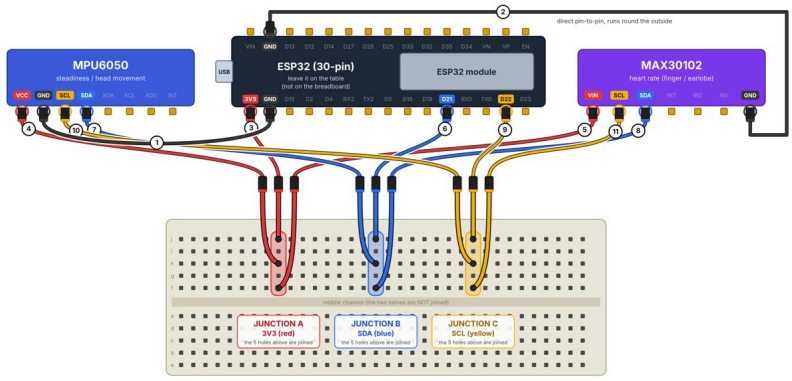


Figure 1: Complete hardware wiring schematic linking the ESP32 (30-pin) to MPU-6050 and MAX30102 via three breadboard junction rails.

Bus Junction / Connection	ESP32 Pin	MPU-6050 Pin	MAX30102 Pin	Functional Description & Electrical Characteristics
Junction A: Power Rail (Red)	3V3	VCC	VIN	Common 3.3V regulated power rail distributed via breadboard column.
Junction B: I²C SDA (Blue)	GPIO21 (D21)	SDA	SDA	Shared bidirectional Serial Data line. Pulled up to 3.3V via 4.7 kΩ resistors.
Junction C: I²C SCL (Yellow)	GPIO22 (D22)	SCL	SCL	Shared Serial Clock line generated by ESP32 master (operated at 100–400 kHz).
Direct Ground Loop (Black)	GND	GND	GND	Direct pin-to-pin perimeter ground routing (wire 2) preventing ground-plane bounce.

> **I²C Addressing & Bus Integrity:** The MPU-6050 defaults to address 0x68 (with AD0 low) while the MAX30102 uses fixed address 0x57. Because their 7-bit addresses are completely distinct, both sensors coexist on the same two physical wires (D21/D22) without requiring multiplexers or software bit-banging.

3. Industrial & Real-Time Telemetry Applications

Fusing cardiovascular biometrics with high-speed inertial measurement enables advanced telemetry solutions across critical industries:

- **Commercial Fleet & Heavy Vehicle Driver Monitoring Systems (DMS):** The MPU-6050 mounted on an operator headset or cap detects characteristic micro-sleep nodding angles (>15° pitch forward for >1.5 seconds) and erratic head roll. Concurrently, the MAX30102 (worn on the earlobe or finger) monitors sudden drops in heart rate variability (HRV) and respiratory rate, triggering audible in-cab alarms and fleet telemetry before micro-sleep causes highway collisions.
- **Aviation & High-G Pilot Physiological Monitoring:** In tactical aviation and unpressurized aerobatics, pilots face gravitational loss of consciousness (G-LOC) and hypobaric hypoxia. The MAX30102 provides rapid real-time blood oxygen saturation (SpO_2) alerts (<90%), while the MPU-6050 measures head steadiness and spatial disorientation against cockpit acceleration vectors.
- **Heavy Machinery, Mining & Crane Operations:** Tower crane operators and open-pit haul truck drivers operate in high-risk, monotonous conditions. Multi-sensor telemetry continuously scores operator alertness, ensuring machine interlocks halt hazardous operations if an operator suffers acute medical distress or falls asleep.
- **Wearable Rehabilitation & Elderly Fall Beacons:** Combines high-impact fall detection (sudden >3g shock followed by stationary horizontal orientation) with instant pulse verification, eliminating false alarms caused by dropped sensor tags.

4. Firmware Driver Architecture & Dual-Sensor Polling

To prevent bus contention and timing jitter on the shared I²C bus, firmware must use an asynchronous, non-blocking polling architecture:

```
// Non-Blocking Dual-Sensor Shared I2C Polling Loop (ESP32)
#include <Wire.h>
#include "MPU6050.h"
#include "MAX30105.h"

MPU6050 imu(0x68);
MAX30105 bioSensor;
unsigned long prevImuMs = 0, prevBioMs = 0;

void setup() {
  Serial.begin(115200);
  Wire.begin(21, 22, 400000); // SDA=21, SCL=22, 400 kHz Fast Mode
  imu.initialize();
  bioSensor.setup(0x1F, 4, 2, 100, 411, 4096); // 100 Hz dual LED mode
}

void loop() {
  unsigned long now = millis();
  // 50 Hz IMU Task: Head Steadiness & Motion Telemetry
  if (now - prevImuMs >= 20) {
    prevImuMs = now;
    int16_t ax, ay, az, gx, gy, gz;
    imu.getMotion6(&ax, &ay, &az, &gx, &gy, &gz);
    // Complementary Filter: Pitch = alpha*(Pitch + gyro*dt) + (1-alpha)*accPitch
  }
  // 25 Hz Biosensor Task: Pulse Rate & Blood Oxygen (SpO2)
  if (now - prevBioMs >= 40) {
    prevBioMs = now;
    long ir = bioSensor.getIR(), red = bioSensor.getRed();
    // SpO2 Derivation: R = (AC_red / DC_red) / (AC_ir / DC_ir)
  }
}
```